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Knowledge-guided Competitive Evolutionary Algorithm for Multi-solution Sensor-Weapon-Target Assignment Problem

Wenhua Li, Xingyi Yao, Kaiwen Li, Rui Wang*, and Tao Zhang

***Abstract*—The Sensor-Weapon-Target Assignment (SWTA) problem in modern air defense systems requires simultaneous optimization of sensor allocation, weapon assignment, and temporal coordination under complex operational constraints. Traditional optimization approaches typically converge to single solutions, failing to capture the multimodal nature of SWTA landscapes where multiple distinct assignment strategies can achieve comparable interception probabilities. This limitation restricts tactical flexibility essential for robust operational planning in dynamic combat environments. To address these challenges, this paper proposes a novel Knowledge-guided Competitive Evolutionary Algorithm (KCEA), integrating with a knowledge extraction mechanism and a competitive selection strategy. The knowledge-guided component systematically identifies and utilizes radar channel conflicts and resource allocation patterns to bias evolutionary search toward promising solution regions, while the competitive selection mechanism ensures preservation of multiple high-quality tactical alternatives throughout the optimization process. Comprehensive experimental validation on benchmark instances demonstrates that KCEA significantly outperforms state-of-the-art methods.**

***Index Terms*—Sensor-weapon-target assignment, multimodal optimization, evolutionary algorithm, knowledge-guided search, competitive selection**

I. Introduction

In the rapidly evolving landscape of modern warfare, air defense systems face unprecedented challenges in countering sophisticated aerial threats. The Sensor-Weapon-Target Assignment (SWTA) problem, which extends the classical Weapon-Target Assignment (WTA) problem by incorporating sensor allocation decisions, represents one of the most critical optimization challenges in military operations research [1]. Unlike traditional WTA formulations that focus solely on weapon-target pairing, SWTA problems require simultaneous optimization of sensor deployment for target detection and tracking, weapon assignment for interception, and temporal coordination of these resources [2].

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The complexity of SWTA problems has intensified dramatically with the emergence of saturation attack tactics, where adversaries deploy large numbers of targets simultaneously to overwhelm defensive capabilities [3]. This tactical evolution necessitates rapid, optimal allocation of limited defensive resources across multiple competing objectives while satisfying stringent temporal and resource constraints, which can be illustrated by Fig. 1. Moreover, the dynamic nature of battlefield environments demands that defensive systems maintain multiple viable solution alternatives, as weapon systems may fail or become unavailable during critical engagement phases.

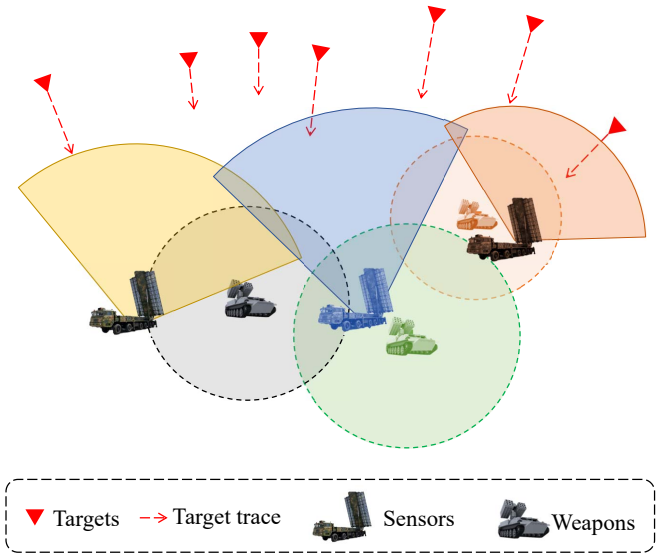


Fig. 1. Illustration of the MS-SWTA problem scenario showing multiple targets (red triangles), sensor detection zones (circular sectors), weapon engagement envelopes (blue circles), and predicted target trajectories (dashed lines). The multimodal nature arises from multiple feasible sensor-weapon-target assignment combinations achieving similar interception probabilities.

A fundamental challenge in SWTA optimization lies in the problem's inherent multimodal nature, where multiple distinct assignment strategies can yield identical or nearly identical interception probabilities [4]. Unlike conventional optimization problems where a single optimal solution suffices, military engagement planning requires multiple alternative strategies for several critical reasons. First, combat situations are highly volatile with constantly changing threat assessments, weapon system failures, and evolving enemy tactics. A single optimal solution may become infeasible within minutes due to equipment malfunctions or updated intelligence. Second,

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different modal solutions often represent distinct tactical approaches, such as concentrated firepower versus distributed engagement patterns. Third, target priority assessments and threat parameters are frequently uncertain or subject to rapid updates during operations, making multiple robust solutions essential for maintaining mission effectiveness under various threat scenarios. Therefore, the multimodal nature of SWTA optimization is not merely a mathematical convenience but a fundamental operational requirement for ensuring mission success in uncertain and dynamic combat environments.

Current approaches to SWTA problems can be broadly categorized into exact methods, heuristic approaches, and metaheuristic algorithms [2]. Exact algorithms can guarantee optimal solutions for small-scale instances but suffer from exponential time complexity that renders them impractical for real-world scenarios involving hundreds of targets and weapons [5]. Heuristic methods, including greedy algorithms and local search techniques [6], offer computational efficiency but often fail to capture the global optimum or maintain solution diversity necessary for tactical flexibility. Metaheuristic algorithms, particularly evolutionary approaches such as Genetic Algorithms (GA) [7], Particle Swarm Optimization (PSO) [8], and Differential Evolution (DE) [9], have shown promising results in addressing large-scale SWTA problems. However, these methods face significant limitations when dealing with the multimodal nature of SWTA landscapes. Traditional evolutionary algorithms typically converge to a single solution region [10], failing to maintain the diverse set of high-quality alternatives that military decision-makers require for robust operational planning.

Furthermore, existing SWTA formulations often oversimplify critical temporal aspects of the problem. Most current models treat interception timing as a derived parameter rather than an optimizable variable, potentially missing opportunities for improved system performance through coordinated timing strategies. The integration of optimal launch timing with sensor-weapon-target assignment decisions represents a significant gap in current literature that directly impacts the practical applicability of proposed solutions.

Recent advances in knowledge-guided optimization have demonstrated the potential for incorporating domain-specific insights to accelerate convergence and improve solution quality in complex optimization problems [11], [12]. However, the application of such knowledge-guided approaches to multimodal SWTA problems remains largely unexplored, particularly in the context of maintaining multiple diverse solutions while leveraging problem-specific information to guide the search process [13].

To address these critical limitations, this paper introduces a novel knowledge-guided competitive evolutionary algorithm specifically designed for multi-solution SWTA optimization, termed KCEA. Unlike existing multimodal optimization methods that rely primarily on fitness-distance balancing for diversity preservation, our knowledge-guided mechanism actively learns from constraint violation patterns to bias the search toward high-quality feasible regions while simultaneously maintaining structural diversity through similarity-aware competition. This synergistic combination creates a positive feedback loop where knowledge extraction benefits from the diverse solution contexts maintained by competitive selection, while the competitive selection process operates on higher-quality solutions generated through knowledge-guided variation. This dual innovation enables KCEA to achieve superior performance in both solution quality and diversity preservation compared to methods that address these objectives independently.

Experimental validation on a comprehensive test suite demonstrates that our approach significantly outperforms existing state-of-the-art methods in terms of both solution quality and diversity maintenance. The algorithm successfully identifies multiple distinct high-quality solutions for large-scale SWTA instances, providing military planners with the tactical flexibility required for robust operational decision-making in dynamic combat environments.

The remainder of this paper is organized as follows. Section II reviews related work in SWTA optimization and multimodal evolutionary algorithms. Section III presents the enhanced SWTA problem formulation incorporating optimal timing decisions. Section IV details the proposed knowledge-guided competitive evolutionary algorithm, including the competitive selection strategy and knowledge extraction mechanisms. Section V provides comprehensive experimental results comparing our approach with existing methods. Finally, Section VI concludes the paper and discusses future research directions.

II. Related Works

A. WTA and SWTA Problem Evolution

The foundational work on weapon-target assignment can be traced to Manne's seminal contribution in 1958 [1], which initially modeled the WTA problem using a nonlinear integer programming approach grounded in probability theory. Comprehensive surveys by Kline et al. [14] and Cai et al. [15] traced the development from Manne's foundational work to contemporary formulations, systematically reviewing various model formulations and highlighting the progression from simple static assignments to complex dynamic scenarios requiring real-time decision-making capabilities.

The transition from pure WTA to integrated SWTA formulations began with recognition that sensor allocation decisions significantly impact overall system performance. Xin et al. [16] built a 0-1 integer programming mathematical model for the co-allocation of sensors and weapons, while Hu et al. [17] introduced dual-threat functions in collaborative aerial warfare contexts. Recent developments have further expanded the problem scope with Lu et al. [18] reformulating WTA as linear integer programming with binary columns, and Pan et al. [19] developing game-theoretic formulations incorporating adversarial strategies. Recent theoretical advances have significantly expanded the scope and complexity of WTA formulations. Bi et al. [20] developed a novel theoretical framework, providing a game-theoretic foundation for UAV weapon-target assignment. Yang et al. [21] introduced a reinforcement learning-based theoretical framework using state split flow deep Q-learning networks, establishing new theoretical connections between threat assessment and optimal

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weapon allocation policies. Tuncer and Cirpan [22] recently established a theoretical framework for networked air defense radar resource management, providing mathematical foundations for collaborative sensor-weapon coordination under power and tracking constraints.

Contemporary research has increasingly focused on multi-objective and multi-stage formulations. Chang et al. [23] investigated multi-stage weapon target assignment across multiple attacking phases, while Wang et al. [24] established constrained multi-objective multi-stage models. Other notable contributions include Liu et al. [25] proposing time-driven dynamic assignment methods, Silav et al. [26] developing bi-objective models for dynamic scenarios, and specialized applications such as Kim et al. [27] addressing heading errors and Bi et al. [28] introducing threshold public goods game theory for UAV assignments.

B. Multimodal Optimization and Algorithmic Advances

The multimodal nature of SWTA problems, where multiple distinct assignment strategies can achieve similar performance levels, necessitates specialized optimization approaches. Classical exact methods by Ahuja et al. [29] developed integer programming and network flow-based approaches within branch-and-bound frameworks, while Andersen et al. [30] introduced linearization techniques with novel "branch-and-adjust" exact solution approaches. Lu et al. [31] presented exact algorithms using column enumeration and branch-and-bound techniques, and Kline et al. [32] improved heuristic approaches based on Quiz Problem solutions.

Recent developments in multimodal optimization [33] have shown promise for combinatorial problems requiring multiple solutions. Yue et al. [34] proposed a hierarchical genetic algorithm for multi-solution traveling salesman problems, implementing a "preprocessing-approximate search-precise search" framework that progressively discovers multiple optimal solutions. Huang et al. [35] developed nLKH-ACS, which combines niching techniques with Lin-Kernighan-Helsgaun heuristics for diversity preservation in large-scale instances. Liu et al. [36], [37] addressed multimodal multiobjective optimization in discrete spaces, developing specialized crossover operators and environmental selection mechanisms to balance objective and decision space diversity. In addition, Li et al. proposed the HREA [38] and CoMMEA [39] to address multimodal multiobjective optimization with local Pareto fronts. Jian and Hsieh [40] introduced niching regression adaptive memetic algorithms specifically designed for large-scale multimodal TSP optimization.

Contemporary metaheuristic developments have addressed large-scale and multi-objective scenarios in defense applications. Hocaoglu et al. [41] used constraint-based nonlinear goal programming for multi-air defense systems, while Yi et al. [42] and Tunga et al. [43] developed multi-objective evolutionary approaches with MOEA/D-iAM2M and comparative MOPSO/NSGA-II analyses respectively. Wang et al. [44] studied SWTA using adaptive large neighborhood search, and Xu et al. [45] developed dynamic Gaussian mutation beetle swarm optimization for large-scale problems. Despite these advances, the application of multimodal optimization techniques to SWTA problems remains limited, particularly for maintaining solution diversity while satisfying complex operational constraints.

C. Knowledge-Guided and Learning-Based Approaches

The integration of domain-specific knowledge into optimization algorithms has emerged as a powerful paradigm for complex constrained problems. As for the theoretical algorithms, Zuo et al. [46] established a comprehensive theoretical framework for process knowledge-guided autonomous evolutionary optimization, introducing formal definitions of knowledge extraction from constraint violation patterns. Yao et al. [47] developed solving knowledge-guided evolutionary algorithms for large-scale many-objective optimization problems, providing mathematical foundations for utilizing solving knowledge to supplement new excellent individuals. Guo et al. [48] contributed to the theoretical understanding of knowledge transfer in dynamic environments, establishing mathematical foundations for evaluating knowledge representativeness and providing theoretical guarantees for positive knowledge transfer.

The practical implementation of knowledge-based optimization has demonstrated effectiveness across diverse domains. Ghosh et al. [49] proposed an interactive framework that extracts variable-wise relationships from high-performing solutions through graph analysis, enabling user feedback integration to steer optimization. Fan et al. [50] developed a differential evolution algorithm utilizing prior knowledge and opposition learning to guide parameter evolution. Song et al. [51] introduced a knowledge-based genetic algorithm for satellite mission scheduling, where domain knowledge guides both initialization and optimization to improve efficiency.

For WTA problems, Xin et al. [52] proposed efficient rule-based constructive heuristics utilizing domain knowledge for direct weapon assignment, demonstrating significant advantages over Monte Carlo approaches. Liu et al. [25] and Peng et al. [53] explored reinforcement learning applications in dynamic weapon target assignment, with the latter addressing multi-ship scenarios using cooperative distributional reinforcement learning. Li et al. [54] provided a comprehensive survey categorizing WTA formulations and solution methods, emphasizing the growing importance of intelligent optimization techniques.

D. Research Gaps

To sum up, despite significant advances, several critical gaps remain: most formulations treat interception timing as derived parameters rather than optimization variables, existing algorithms fail to maintain diverse high-quality alternatives required for tactical flexibility, and constraint violation information is largely underutilized during optimization.

III. Problem Formulation

This section presents a comprehensive mathematical formulation of the Multi-Solution Sensor-Weapon-Target Assignment (MS-SWTA) problem, which extends traditional SWTA

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formulations by explicitly considering optimal interception timing and the inherent multimodal nature of solution landscapes.

A. Problem Description and Assumptions

Consider an air defense scenario (shown in Fig. 1) where a set of incoming aerial targets $\mathcal{T} = \{T_1, T_2, \ldots, T_{N_T}\}$ must be intercepted using available defensive resources comprising sensor platforms $\mathcal{S} = \{S_1, S_2, \ldots, S_{N_S}\}$ and weapon systems $\mathcal{W} = \{W_1, W_2, \ldots, W_{N_W}\}$. Each target follows a predictable trajectory within a discrete time horizon $\mathcal{H} = \{t_1, t_2, \ldots, t_{N_H}\}$, where effective interception must occur before target impact.

To ensure mathematical rigor and clarity, we establish a comprehensive notation system that captures the complex interdependencies inherent in the MS-SWTA problem, which is listed in TABLE I.

The multi-solution aspect of SWTA is particularly important for operational flexibility and risk mitigation. Each modal solution typically represents a different tactical approach to threat engagement, such as: (1) Concentrated engagement strategies that focus maximum firepower on high-priority targets, (2) Distributed engagement strategies that spread defensive resources across multiple threat vectors, and (3) Layered defense strategies that sequence engagements based on target approach timing. These diverse solution modes provide commanders with contingency options that remain effective even when primary solutions become compromised due to equipment failures, updated threat intelligence, or unexpected enemy maneuvers. The availability of multiple high-quality alternatives significantly enhances the robustness and adaptability of defensive operations.

B. Mathematical Model Formulation

1) Objective Function: The primary objective of the MS-SWTA problem is to maximize the overall system interception probability while maintaining solution diversity to support tactical flexibility. The objective formulation is expressed as:

$$\max_{\mathbf{X}\in\Omega} \quad F(\mathbf{X}) = \sum_{i=1}^{N_T} V_i \cdot P_i^{\text{total}}(\mathbf{X}) \tag{1}$$

where the total interception probability for target T_i follows the probabilistic independence assumption:

$$P_i^{\text{total}}(\mathbf{X}) = 1 - \prod_{j\in\mathcal{K}_i} \left(1 - \gamma_i^j \cdot P_i^j(\mathbf{X})\right) \tag{2}$$

The individual interception probability $P_i^j(\mathbf{X})$ depends on the sensor-weapon pairing and optimal timing:

$$P_i^j(\mathbf{X}) = \sum_{s\in\mathcal{S}} \sum_{w\in\mathcal{W}} \omega_{iw}^j \cdot \sigma_{is}^{jk^*} \cdot P_{\text{kill}}(d_{iw}(\tau_i^j), \text{ACC}_{is}) \tag{3}$$

where $k^* = \arg\min_k |t_k - \tau_i^j|$ represents the discrete time closest to the optimal launch time, and P_{kill} is the kill probability function based on miss distance d_{iw} and sensor accuracy ACC_{is}.

2) Constraint Formulation: The MS-SWTA problem involves several categories of constraints that ensure operational feasibility and resource coordination.

Assignment Consistency Constraints: These constraints ensure each target interception attempt is assigned to at most one sensor-weapon pair and maintain consistency between assignment variables:

$$\sum_{s\in\mathcal{S}} \sum_{w\in\mathcal{W}} x_{isw}^{jk} \le 1, \quad \forall i \in \mathcal{T}, j \in \mathcal{K}_i, k \in \mathcal{H} \tag{4}$$

$$\sum_{k\in\mathcal{H}} x_{isw}^{jk} = \omega_{iw}^j \cdot \sigma_{is}^{jk}, \quad \forall i, s, w, j \tag{5}$$

Sensor Capacity Constraints: The number of simultaneous tracking channels is limited and targets are visible to the sensors (see Fig. 2):

$$\sum_{i\in\mathcal{T}} \sum_{j\in\mathcal{K}_i} \sigma_{is}^{jk} \le C_s^{\max}, \quad \forall s \in \mathcal{S}, k \in \mathcal{H} \tag{6}$$

$$\sigma_{is}^{jk} \le \alpha_{is}(t_k), \quad \forall i, s, j, k \tag{7}$$

Weapon Resource Constraints: These constraints manage missile inventory limits and minimum reload times between successive launches:

$$\sum_{i\in\mathcal{T}} \sum_{j\in\mathcal{K}_i} \omega_{iw}^j \le M_w^{\text{load}}, \quad \forall w \in \mathcal{W} \tag{8}$$

$$|\tau_i^j - \tau_i^{j'}| \ge \Delta t_w^{\text{reload}}, \quad \forall i, j \ne j', w : \omega_{iw}^j = \omega_{iw}^{j'} = 1 \tag{9}$$

Temporal Constraints: These constraints ensure proper timing coordination between tracking initiation, missile launch, and target interception:

$$\tau_i^j + t_{\text{flight}}^{iw} \le T_i^{\text{impact}}, \quad \forall i, j, w : \omega_{iw}^j = 1 \tag{10}$$

$$\tau_i^j \ge t_{\text{track}}^{is} + t_{\text{lock}}^{is}, \quad \forall i, j, s : \sigma_{is}^{jk} = 1 \tag{11}$$

Engagement Envelope Constraints: Weapons are only assigned to targets within their kinematic engagement capability:

$$\omega_{iw}^j \le \beta_{iw}(\tau_i^j), \quad \forall i, j, w \tag{12}$$

C. Multimodal Solution Landscape

A critical characteristic of the MS-SWTA problem is its inherent multimodal nature, where multiple distinct assignment strategies can achieve identical or nearly identical interception probabilities. This multimodality arises from several sources: 1) Multiple sensor-weapon combinations may provide equivalent interception capabilities for specific targets. 2) Different launch timing strategies can yield similar overall performance through compensation effects. 3) Various resource allocation patterns may achieve similar system-wide utilization while maintaining comparable effectiveness. Formally, the solution space Ω contains multiple regions $\Omega_1, \Omega_2, \ldots, \Omega_M$ such that:

$$\max_{\mathbf{X}\in\Omega_m} F_1(\mathbf{X}) \approx \max_{\mathbf{X}\in\Omega_{m'}} F_1(\mathbf{X}), \quad m \ne m' \tag{13}$$

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TABLE I
MATHEMATICAL NOTATION AND SYMBOL DEFINITIONS FOR THE MS-SWTA PROBLEM FORMULATION.

Category	Symbol	Definition
Problem Sets	$\mathcal{T}$	Set of incoming targets, $\lvert\mathcal{T}\rvert = N_T$
	$\mathcal{S}$	Set of available sensors, $\lvert\mathcal{S}\rvert = N_S$
	$\mathcal{W}$	Set of weapon systems, $\lvert\mathcal{W}\rvert = N_W$
	$\mathcal{H}$	Discrete time horizon, $\lvert\mathcal{H}\rvert = N_H$
	$\mathcal{K}_i$	Set of possible interception attempts for target T_i, $\lvert\mathcal{K}_i\rvert \leq K_{\max}$
	$\mathcal{F}_{ij}$	Set of feasible sensor-weapon pairs for target T_i at attempt j
Target Parameters	$\mathbf{p}_i(t)$	Position vector of target T_i at time t
	v_i	Velocity magnitude of target T_i
	θ_i	Heading angle of target T_i
	V_i	Strategic value/importance weight of target T_i
	T_i^{impact}	Predicted impact time for target T_i
Sensor Parameters	R_s^{det}	Detection range of sensor S_s
	R_s^{track}	Tracking range of sensor S_s
	$C_s^{\max}$	Maximum tracking channel capacity of sensor S_s
	$\alpha_{is}(t)$	Binary visibility indicator: 1 if sensor S_s can detect target T_i at time t, 0 otherwise
Weapon Parameters	R_w^{eng}	Engagement range of weapon system W_w
	M_w^{load}	Missile inventory of weapon system W_w
	$\Delta t_w^{\text{reload}}$	Minimum reload time between successive launches from W_w
	$\beta_{iw}(t)$	Binary engagement indicator: 1 if weapon W_w can engage target T_i at time t, 0 otherwise
Decision Variables	x_{isw}^{jk}	Binary assignment: 1 if target T_i, attempt j is assigned to sensor S_s and weapon W_w at time t_k, 0 otherwise
	τ_i^j	Optimal launch time for j-th interception attempt on target T_i
	σ_{is}^{jk}	Binary sensor allocation: 1 if sensor S_s tracks target T_i during attempt j at time t_k, 0 otherwise
	ω_{iw}^j	Binary weapon allocation: 1 if weapon W_w is assigned to target T_i for attempt j, 0 otherwise
	γ_i^j	Binary attempt activation: 1 if j-th interception attempt on target T_i is executed, 0 otherwise
	$\mathbf{X}$	Complete solution vector comprising all decision variables
Performance Metrics	$P_i^j(\mathbf{X})$	Interception probability for target T_i at attempt j under solution $\mathbf{X}$
	$P_i^{\text{total}}(\mathbf{X})$	Total interception probability for target T_i under solution $\mathbf{X}$
	$P^{\text{system}}(\mathbf{X})$	Overall system interception probability under solution $\mathbf{X}$
	$C^{\text{resource}}(\mathbf{X})$	Total resource utilization cost under solution $\mathbf{X}$
	$d(\mathbf{X}_1, \mathbf{X}_2)$	Similarity distance between solutions $\mathbf{X}_1$ and $\mathbf{X}_2$

while the solutions within different regions exhibit significant structural differences:

$$d(\mathbf{X}_m, \mathbf{X}_{m'}) > \delta_{\text{threshold}}, \quad \mathbf{X}_m \in \Omega_m, \mathbf{X}_{m'} \in \Omega_{m'} \tag{14}$$

This multimodal landscape presents both opportunities and challenges. While it provides tactical flexibility and robustness against dynamic battlefield conditions, it complicates optimization by requiring algorithms to maintain diverse populations and avoid premature convergence to single solution regions.

The extensive solution diversity generated by the algorithm possesses significant practical implications that extend beyond algorithmic performance enhancement to directly strengthen battlefield resilience and operational adaptability. When weapon system failures occur, commanders can immediately select alternative solutions from pre-computed solution sets to redistribute interception responsibilities. For instance, when Solution **A**'s reliance on long-range interceptors becomes compromised due to system unavailability, Solution **B** emphasizing coordinated short-range engagement can be promptly implemented. Simultaneously, when target trajectory predictions are updated based on new intelligence, the diversified solution portfolio enables rapid selection of engagement strategies optimized for different trajectory assumptions, where conservative solutions maintain effectiveness across broader uncertainty ranges while specialized solutions optimize performance for specific threat scenarios.

IV. KNOWLEDGE-GUIDED COMPETITIVE EVOLUTIONARY ALGORITHM

To effectively address the multimodal nature of MS-SWTA problems while maintaining computational efficiency, this section presents a novel Knowledge-Guided Competitive Evolu-

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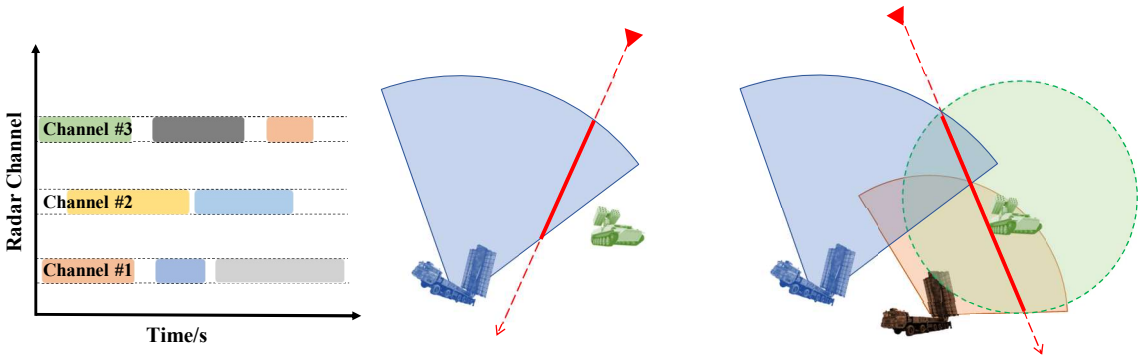


Fig. 2. Illustration of radar related constraints. For each radar, the total channel is limited. In addition, the target can only be intercepted within the feasible range of radar and missile.

tionary Algorithm (KCEA). The algorithm integrates domain-specific knowledge extraction with a competitive selection mechanism designed to preserve solution diversity across multiple high-quality regions of the search space.

A. Algorithm Framework

The overall framework of KCEA is designed to simultaneously optimize solution quality and maintain population diversity throughout the evolutionary process. Algorithm 1 presents the main algorithmic structure, which integrates knowledge-guided variation operators with competitive environmental selection to effectively navigate the multimodal MS-SWTA landscape.

Algorithm 1 Knowledge-Guided Competitive Evolutionary Algorithm (KCEA)

Require: Population size N, maximum generations G_{max}, problem instance $\mathcal{P}$
Ensure: Diverse solution set $\mathcal{D} = \{\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_M\}$
$\mathcal{P}(0) \leftarrow$ Initialize(N, $\mathcal{P}$) {Random population initialization}
$\mathcal{K}(0) \leftarrow \emptyset$ {Initialize knowledge base}
$t \leftarrow 0$ {Generation counter}
while $t < G_{max}$ and not converged **do**
  $\mathcal{K}(t) \leftarrow$ ExtractKnowledge($\mathcal{P}(t)$) {Extract conflict patterns}
  $\mathcal{Q}(t) \leftarrow$ KnowledgeGuidedVariation($\mathcal{P}(t)$, $\mathcal{K}(t)$) {Generate offspring}
  Evaluate($\mathcal{Q}(t)$) {Fitness evaluation}
  $\mathcal{P}(t+1) \leftarrow$ CompetitiveSelection($\mathcal{P}(t) \cup \mathcal{Q}(t)$, N) {Environmental selection}
  UpdateDiverseSet($\mathcal{D}$, $\mathcal{P}(t+1)$) {Maintain diverse solutions}
  $t \leftarrow t + 1$
end while
return $\mathcal{D}$

The initialization phase creates a random population $\mathcal{P}(0)$ of size N using the encoding scheme described in Section IV-B. During each generation, the knowledge extraction module analyzes the current population to identify recurring conflict patterns, resource bottlenecks, and successful assignment strategies. This extracted knowledge $\mathcal{K}(t)$ is then integrated into the variation operators to bias the search toward regions with higher probability of generating feasible, high-quality solutions. In addition, the competitive selection mechanism represents the core innovation of KCEA for maintaining solution diversity. Instead of traditional fitness-based selection, this mechanism implements similarity-aware competition where solutions compete primarily against similar alternatives.

The external diverse solution set $\mathcal{D}$ is continuously updated throughout the evolutionary process using a similarity-based clustering approach. This ensures that the final output represents multiple distinct tactical alternatives, each corresponding to different regions of the multimodal solution landscape.

The KCEA framework incorporates several key innovations that distinguish it from conventional evolutionary algorithms: 1) Unlike traditional EAs that rely solely on random variation operators, KCEA extracts domain-specific knowledge from the constraint violation patterns observed during the evolutionary process. 2) The environmental selection process employs a competitive strategy that explicitly considers solution similarity alongside fitness values. 3) The algorithm maintains an external archive $\mathcal{D}$ of diverse high-quality solutions that spans different regions of the multimodal landscape.

B. Solution Encoding and Representation

The MS-SWTA problem exhibits a natural bi-level optimization structure where the upper level determines sensor-weapon pair assignments and the lower level optimizes launch timing under these assignments. In this study, we propose a dual-layer encoding scheme that efficiently represents both assignment decisions and extracted knowledge from constraint violations, which can be illustrated by Fig. 3.

First Layer: Each target T_i is associated with a sequence of integers corresponding to multiple interception attempts. Each integer represents the index of a pre-computed feasible sensor-weapon pair from the enumerated set $\mathcal{F}_i$. For target T_i with up to $\mathcal{K}_{\max}$ interception attempts, the encoding is represented as $[a_i^1, a_i^2, \ldots, a_i^{\mathcal{K}_{\max}}]$ where $a_i^j \in \{0, 1, \ldots, |\mathcal{F}_i|\}$ and $a_i^j = 0$ indicates no assignment for attempt j.

Second Layer: This layer maintains extracted conflict information in the form of target pairs and attempt indices that experience resource conflicts during decoding. The knowledge

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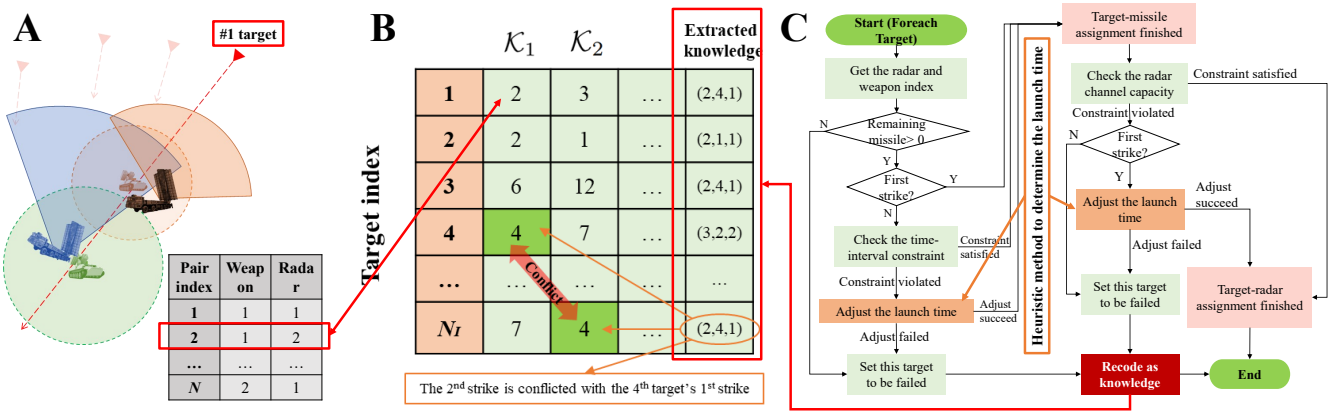


Fig. 3. Dual-layer encoding-decoding scheme for MS-SWTA solutions. **(A)** Before the optimization, the feasible pairs of (weapon,radar) for each target are calculated. **(B)** The dual-layer decision variable for MS-SWTA. For each strike of each target, the index of (weapon,radar) is recorded, where the conflict information is represented as a tuple. **(C)** The decoding method for calculating the objective value for MS-SWTA, where the best launch time is calculated by a heuristic method.

is represented as conflict tuples (j_1, T_i, j_2) indicating that the j_1-th attempt conflicts with the j_2-th attempt on target T_i due to radar channel or weapon resource limitations.

The lower level timing optimization is addressed through a heuristic decoding process which follows a priority-based approach where targets are ordered by strategic importance and time criticality. For each active assignment, the algorithm calculates the optimal launch time τ_i^j by minimizing the miss distance while ensuring sufficient tracking time and respecting weapon reload constraints. When resource conflicts arise, the algorithm attempts temporal adjustments within the feasible launch window. If conflicts cannot be resolved, the assignment is deactivated and the conflict information is recorded in the knowledge repository. This approach effectively transforms the bi-level optimization problem into a single-level assignment optimization with embedded timing heuristics.

C. Knowledge Extraction and Utilization

The knowledge extraction and utilization mechanism forms a core component of KCEA that learns from constraint violations encountered during solution decoding to guide subsequent evolutionary operations. This approach transforms typically discarded infeasibility information into valuable domain knowledge that accelerates convergence toward feasible and high-quality solution regions.

1) Conflict Pattern Identification: During the solution decoding process described in Section IV-B, the algorithm systematically identifies three primary types of resource conflicts. Radar channel conflicts occur when multiple target tracking requirements exceed sensor capacity limits at specific time intervals. Weapon resource conflicts arise from missile inventory limitations or reload time violations when the same launcher is assigned to multiple targets with insufficient temporal separation. Temporal coordination conflicts emerge when optimal launch timing violates tracking prerequisites or interception window constraints.

Each conflict is characterized by the involved target-attempt pairs and the temporal context of the violation. The algorithm records this information as conflict tuples. This structured representation enables systematic analysis of conflict patterns and their utilization in guiding evolutionary search.

2) Knowledge-Guided Variation Operators: The extracted conflict knowledge directly influences both crossover and mutation operations to bias the search toward conflict-free solution regions. As illustrated in Fig. 4, the knowledge-guided operators selectively target problematic solution components identified through conflict analysis.

Knowledge-Guided Crossover: The crossover operator uses conflict information to determine which solution segments to exchange between parent solutions. Targets frequently involved in conflicts receive higher selection probabilities for segment exchange, allowing the algorithm to explore alternative assignments for problematic targets while preserving successful assignment patterns for conflict-free targets. The crossover process calculates selection probabilities p_i for each target T_i based on its conflict frequency: $p_i = 1 + c_i$ where c_i is the number of conflicts involving target T_i. Algorithm 2 presents the complete procedure.

Knowledge-Guided Mutation: The mutation operator implements two complementary strategies guided by conflict knowledge. Single-gene mutation directly modifies assignments for targets with high conflict involvement, replacing problematic sensor-weapon pairs with alternative feasible combinations. Attempt-exchange mutation reorders interception attempts for the same target to resolve temporal coordination conflicts while maintaining resource assignments.

3) Knowledge-Enhanced Environmental Selection: Beyond guiding variation operators, the extracted knowledge also influences environmental selection decisions when comparing solutions with identical objective values. The algorithm employs a conflict-aware comparison mechanism that favors solutions with fewer active conflicts, even when objective function values are equivalent. This preference promotes feasibility and helps maintain population movement toward constraint-satisfied regions.

The knowledge utilization mechanism creates a feedback loop between constraint violation detection and evolutionary

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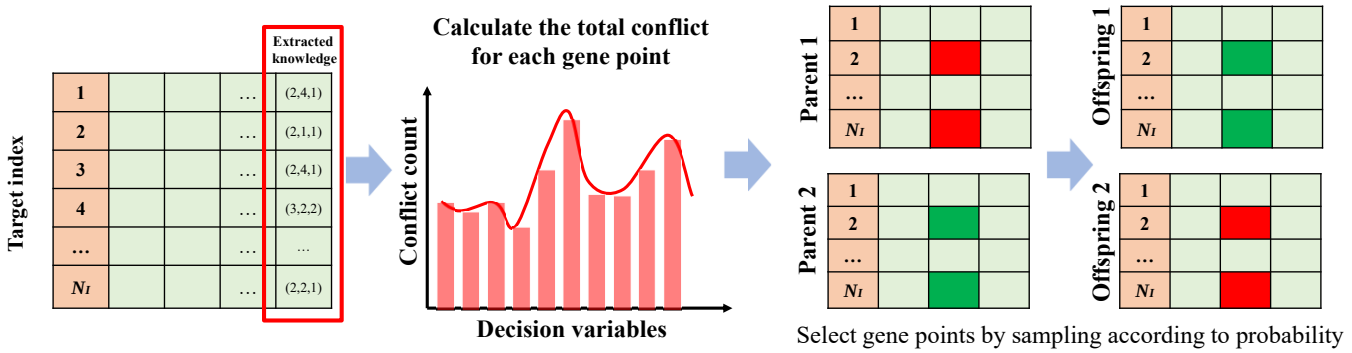


Fig. 4. Knowledge-guided variation operators utilizing extracted conflict information. The example shows how radar channel conflicts between target pairs (e.g., T_2 and T_{N_I} competing for radar S_1) are identified and recorded as pairwise conflict tuples, which then guide crossover and mutation operations to explore alternative assignments that avoid similar conflicts.

Algorithm 2 Knowledge-Guided Crossover Operator

Require: Parent solutions $\mathbf{X}_1$, $\mathbf{X}_2$, conflict knowledge $\mathcal{K}$
Ensure: Offspring solutions $\mathbf{Y}_1$, $\mathbf{Y}_2$
// Calculate conflict-based selection probabilities
for $i = 1$ to N_T **do**
  $c_i \leftarrow$ CountConflicts(T_i, $\mathcal{K}$) {Count conflicts involving target T_i}
  $p_i \leftarrow \frac{1+c_i}{\sum_{j=1}^{N_T}(1+c_j)}$ {Normalized selection probability}
end for
// Determine crossover points using conflict probabilities
$\mathcal{C} \leftarrow \emptyset$ {Set of selected crossover targets}
for $i = 1$ to N_T **do**
  if Random(0,1) $< p_i$ **then**
    $\mathcal{C} \leftarrow \mathcal{C} \cup \{T_i\}$
  end if
end for
// Perform segment exchange
$\mathbf{Y}_1 \leftarrow \mathbf{X}_1$, $\mathbf{Y}_2 \leftarrow \mathbf{X}_2$
for $T_i \in \mathcal{C}$ **do**
  Exchange assignment segments for T_i between $\mathbf{Y}_1$ and $\mathbf{Y}_2$
end for
return $\mathbf{Y}_1$, $\mathbf{Y}_2$

search guidance, effectively transforming the typically wasteful exploration of infeasible regions into a learning opportunity that accelerates convergence. This approach is particularly valuable for highly constrained problems like MS-SWTA where feasible solutions occupy a small fraction of the total search space.

4) Illustrative Example of Knowledge Extraction: To illustrate the knowledge extraction mechanism, consider a scenario where radar S_1 has a maximum channel capacity of $C_1^{\max} = 2$ and is already tracking targets T_1 and T_2 at time t_k. During the decoding process, the algorithm attempts to assign target T_{N_I} to radar S_1 for its first interception attempt. This creates a radar channel conflict since S_1 cannot simultaneously track three targets.

The knowledge extraction mechanism records this conflict as a tuple: $T_{N_I} : (2, T_2, 1)$, indicating that the second attempt on target T_{N_I} conflicts with the first attempt on target T_2 due to radar resource limitations. The algorithm identifies that T_2 was the last target assigned to S_1 before the capacity limit was exceeded, making it the direct conflicting target with T_{N_I}. In subsequent generations, this extracted knowledge influences the evolutionary operators as follows:

Knowledge-Guided Crossover: Both targets T_2 and T_{N_I} receive elevated selection probabilities for crossover operations, with $p_2 = p_3 = 1 + 1 = 2$, making them more likely to be selected for segment exchange with other parent solutions that may have different radar assignments.

Knowledge-Guided Mutation: The mutation operator specifically targets the problematic assignment of T_{N_I} to (S_1, W_1), replacing it with an alternative feasible pairing such as (S_2, W_2), while preserving T_2's assignment to maintain solution stability.

This targeted modification resolves the radar channel conflict while maintaining high interception probabilities. The algorithm learns that assigning T_2 and T_{N_I} to the same radar simultaneously should be avoided, and this pairwise conflict knowledge continues to guide future evolutionary operations, preventing similar constraint violations and accelerating convergence toward feasible solutions.

D. Competitive Selection Strategy

Unlike traditional fitness-based selection that typically converges to single solution regions, this mechanism implements similarity-aware competition to preserve multiple high-quality solutions across different modal regions while gradually improving overall population fitness.

1) Solution Similarity Measurement: Effective diversity preservation requires accurate measurement of structural differences between MS-SWTA solutions. The algorithm employs a multi-component similarity metric that captures both assignment pattern differences and temporal coordination variations. For two solutions $\mathbf{X}_1$ and $\mathbf{X}_2$, the similarity distance is computed as:

$$d(\mathbf{X}_1, \mathbf{X}_2) = \frac{\sum_{i=1}^{N_T} \sum_{j=1}^{\mathcal{K}_{\max}} \mathbf{1}[a_i^j(\mathbf{X}_1) \neq a_i^j(\mathbf{X}_2)]}{N_T \cdot \mathcal{K}_{\max}} \quad (15)$$

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where $\mathbf{1}[\cdot]$ means the total number that satisfy the condition. This similarity metric effectively distinguishes between solutions that represent genuinely different tactical approaches versus minor variations of the same basic strategy.

2) Similarity-Based Competition Mechanism: The competitive selection mechanism implements a tournament-based approach where solutions compete primarily against similar alternatives rather than the entire population. The competition process randomly selects one solution from the combined parent-offspring population and identifies its most similar competitor using the distance metric defined above. These two solutions then engage in direct comparison where the winner is determined by fitness values when differences exceed a threshold ϵ, or by constraint violation counts when fitness values are comparable. This localized competition ensures that each solution type faces appropriate selection pressure without being dominated by potentially superior but structurally different alternatives. Algorithm 3 outlines the competitive selection procedure.

Algorithm 3 Competitive Selection Strategy

Require: Combined population $\mathcal{P}_{\text{combined}}$, population size N
Ensure: Next generation population $\mathcal{P}_{\text{next}}$

```
P_next ← ∅
while |P_next| < N do
    X_1 ← RandomSelect(P_combined)
    X_2 ← FindMostSimilar(X_1, P_combined \ {X_1})
    X_winner ← CompetitiveTournament(X_1, X_2)
    P_next ← P_next ∪ {X_winner}
    P_combined ← P_combined \ {X_1, X_2}
end while
return P_next
```

Unlike traditional fitness-based selection that converges rapidly to the best-discovered region, the competitive strategy maintains multiple solution clusters simultaneously by ensuring solutions compete primarily within their similarity neighborhoods. This approach enables discovery and preservation of multiple distinct tactical strategies with comparable effectiveness but fundamentally different resource allocation approaches. For instance, one modal region might emphasize early interception using long-range weapons, while another focuses on coordinated multi-attempt strategies using shorter-range systems. Both approaches can be tactically valuable depending on operational circumstances.

E. Time Complexity Analysis

The computational complexity of KCEA consists of several key components that determine its overall time complexity and real-time applicability, includes: (1) Population evaluation: Each individual requires constraint checking and fitness calculation with complexity $O(T \cdot S)$ for assignment validation and $O(T \cdot K)$ for interception probability computation, where K is the maximum number of interception attempts per target. For the entire population, this yields $O(N \cdot T \cdot S)$. (2) Knowledge extraction: Conflict detection involves checking resource constraints for all assignments, requiring $O(T \cdot S)$ per individual. Across the population, knowledge extraction complexity is $O(N \cdot T \cdot S)$. (3) Knowledge-guided crossover and mutation: Parent selection using conflict-based probabilities requires $O(N)$ operations. The crossover operation itself involves segment exchange with complexity $O(T)$ per offspring generation, resulting in $O(N \cdot T)$ for producing N offspring. (4) Competitive selection: Similarity computation between individuals requires $O(T)$ operations per pair, resulting in $O(N^2 \cdot T)$ for complete similarity matrix construction. Neighborhood-based competition adds $O(N^2)$ complexity for selection operations.

In conclusion, the dominant terms are the competitive selection phase $O(N^2 \cdot T)$ and mutation operations $O(N \cdot T \cdot S^2)$, making the overall complexity $O(N^2 \cdot T + N \cdot T \cdot S^2)$ per generation.

V. Experiment

A. Experimental Setup

1) Baseline Algorithms: To comprehensively evaluate the performance of the proposed KCEA for MS-SWTA optimization, we conduct comparative analysis against five state-of-the-art algorithms representing different algorithmic paradigms: Heuristic Algorithm (HA), Adaptive Large-scale Neighborhood Search (ALNS) [6], Genetic Algorithm for SWTA (GA-SWTA) [55], Niching Memetic Algorithm for Multi-Solution Traveling Salesman Problem (NMA) [56] and Fitness and Distance-based Local Search with Adaptive Differential Evolution (FDLS-ADE) [57].

The HA serves as a deterministic baseline that implements a greedy assignment strategy based on Euclidean distance metrics and resource availability, followed by conflict resolution through temporal adjustment heuristics. ALNS represents the metaheuristic approach, employing adaptive destroy-and-repair operators to explore solution neighborhoods systematically. GA-SWTA implements a specialized genetic algorithm with multi-layer encoding and simulation-based decoding specifically designed for SWTA problems. NMA is selected as a representative multimodal optimizer for discrete problem. For NMA, we retain the clearing procedure and crowding-based selection while replacing TSP-specific local search for SWTA constraints. FDLS-ADE is a dedicated multimodal optimization algorithm that maintain solution diversity through fitness-distance balancing strategies, which is also a recently proposed representative algorithm. Specifically, the differential evolution mutation is replaced with gene-position recombination.

In order to comprehensively evaluate the performance of KCEA, we use the average fitness and the average number of different optimal solutions as the performance metrics. To ensure fair comparison, all algorithms adopt the same encoding method. The population size is set to $N_{\text{pop}} = \min\{2 \cdot N_T, 500\}$, where N_T represents the number of targets. The maximum function evaluations are constrained to $\text{FE}_{\max} = 1000 \cdot N_T$ to maintain computational budget consistency. All populations undergo random initialization to eliminate initialization bias. Statistical significance is ensured through 30 independent runs per algorithm. The experimental platform consists of an Intel i9-9900X processor operating at 3.50 GHz with 64 GB RAM.

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2) Benchmark Test Suite: The experimental evaluation employs the MADRA-LIB [13] benchmark suite, which provides realistic representations of Multi-missile Air Defense Resource Allocation Problems across diverse operational scenarios. This benchmark incorporates varying levels of complexity through different target densities, resource constraints, and tactical configurations that reflect real-world air defense challenges. The detailed characteristics of the ten test instances are presented in TABLE II, spanning from small-scale scenarios with 10 targets to large-scale operations involving 500 targets. In addition, we only consider the situation with single and double round interception.

TABLE II
CHARACTERISTICS OF TEST INSTANCES IN MADRA-LIB BENCHMARK SUITE

Instance	N_T	N_S	N_W	M_w^{load}	$C_s^{\max}$
P1	10	4	4	3	3
P2	20	5	5	4	4
P3	24	6	6	5	5
P4	24	6	6	7	7
P5	50	8	8	9	9
P6	50	8	8	10	11
P7	100	10	10	12	12
P8	100	10	10	15	15
P9	200	15	15	16	18
P10	500	20	20	28	28

The benchmark progression systematically increases problem complexity through multiple dimensions: target quantity ranges from 10 to 500, representing scenarios from localized skirmishes to large-scale coordinated attacks; resource availability scales proportionally to maintain realistic defense-to-threat ratios; and constraint tightness varies through different missile load and radar channel configurations. This comprehensive test suite enables thorough evaluation of algorithmic performance across the full spectrum of operational scenarios encountered in modern air defense applications.

B. Result Analysis

1) Performance on the benchmark problems: Table III presents the comprehensive experimental results comparing KCEA against four baseline algorithms across ten benchmark instances, measuring both single and double interception scenarios. The results demonstrate the superior performance of KCEA in terms of solution quality, particularly as problem complexity increases.

For small-scale instances, all evolutionary algorithms achieve identical or nearly identical optimal solutions, indicating that these problems are relatively tractable and multiple approaches can find high-quality solutions. However, the deterministic HA baseline consistently produces slightly suboptimal results, particularly evident in the double interception scenarios where coordinated timing becomes critical for maximizing overall system effectiveness.

As problem complexity increases to medium-scale instances (**P5-P6**), performance differentiation emerges. KCEA achieves optimal solutions of 40.000 in single scenarios, outperforming NMA, FDLS-ADE, GA-SWTA, and HA. The advantages become more substantial in double interception scenarios, where KCEA achieves approximately 45.4 compared to roughly 43.0 for NMA and FDLS-ADE.

Performance advantages become most pronounced in large-scale instances. For **P10** with 500 targets, KCEA reaches 403.890 in double scenarios, significantly exceeding NMA and FDLS-ADE by 7-8 units. Notably, NMA performs comparably to FDLS-ADE on smaller problems but shows greater performance degradation on larger instances, suggesting discrete-native niching mechanisms face scalability challenges.

The knowledge extraction mechanism's effectiveness becomes particularly evident in complex scenarios. Unlike traditional evolutionary algorithms that may waste computational resources exploring infeasible regions, KCEA systematically learns from constraint violations to bias subsequent search operations. This learning capability manifests as progressively improved performance on larger instances, where resource conflicts and temporal coordination requirements create intricate constraint landscapes that challenge conventional optimization approaches. KCEA's consistent superiority against both adapted continuous (FDLS-ADE) and discrete-native multimodal (NMA) methods validates the synergistic value of integrating domain-specific knowledge with diversity preservation.

2) Multi-solutions for SWTA: Table IV presents the number of unique optimal solutions identified by each algorithm across 30 independent runs, providing critical insights into diversity preservation capabilities essential for tactical flexibility in air defense operations.

KCEA demonstrates exceptional performance in identifying multiple distinct optimal solutions, consistently outperforming all baselines across all problem instances. This superior diversity preservation directly addresses tactical requirements for maintaining alternative operational strategies in dynamic combat environments.

The deterministic nature of HA limits its diversity to single solutions per scenario, as expected from a greedy heuristic approach. Interestingly, HA fails to find known optimal solutions for double interception scenarios in medium-scale problems (**P5**-**P6**) and cannot identify solutions for large-scale instances (**P7**-**P8**). ALNS demonstrates moderate diversity preservation capabilities, identifying up to 12 distinct solutions for medium-scale single interception scenarios. However, its performance degrades significantly as problem complexity increases, particularly in double interception scenarios where coordination requirements intensify. GA-SWTA exhibits limited diversity preservation, typically finding fewer distinct solutions than ALNS despite employing population-based search mechanisms. This performance indicates that traditional genetic operators without specialized diversity preservation mechanisms are insufficient for maintaining solution variety in multimodal MS-SWTA landscapes.

Among multimodal optimization methods, NMA and FDLS-ADE exhibit scale-dependent performance patterns. For medium-scale problems, NMA slightly outperforms FDLS-ADE in diversity preservation (e.g., **P5**: 22 vs. 25 distinct so-

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TABLE III
AVERAGE AND STANDARD DEVIATION OF TOTAL INTERCEPTION PROBABILITY OBTAINED BY ALL COMPARED ALGORITHMS OVER 30 INDEPENDENT RUNS, WHERE THE BEST RESULTS FOR EACH TEST INSTANCE ARE HIGHLIGHTED WITH BOLD FONT.

Problems	N_T	HA		ALNS		GA-SWTA		NMA		FDLS-ADE		KCEA	
		Single	Double	Single	Double	Single	Double	Single	Double	Single	Double	Single	Double
P1	10	8.000 ±0.000	9.100 ±0.000	8.000 ±0.000	9.100 ±0.000	8.000 ±0.000	9.100 ±0.000	8.000 ±0.000	9.100 ±0.000	8.000 ±0.000	9.100 ±0.000	**8.000** ±0.000	**9.100** ±0.000
P2	20	16.000 ±0.000	18.200 ±0.000	16.000 ±0.000	18.200 ±0.000	16.000 ±0.000	18.200 ±0.000	16.000 ±0.000	18.200 ±0.000	16.000 ±0.000	18.200 ±0.000	**16.000** ±0.000	**18.200** ±0.000
P3	24	19.200 ±0.000	21.556 ±0.000	19.200 ±0.000	21.680 ±0.125	19.200 ±0.000	21.720 ±0.089	19.200 ±0.000	21.780 ±0.065	19.200 ±0.000	21.750 ±0.075	**19.200** ±0.000	**21.840** ±0.000
P4	24	19.200 ±0.000	21.499 ±0.000	19.200 ±0.000	21.685 ±0.142	19.200 ±0.000	21.745 ±0.098	19.200 ±0.000	21.765 ±0.085	19.200 ±0.000	21.780 ±0.068	**19.200** ±0.000	**21.840** ±0.000
P5	50	35.653 ±0.000	38.634 ±0.000	37.245 ±0.845	41.256 ±1.245	38.127 ±0.652	42.358 ±0.985	38.456 ±0.568	43.125 ±0.856	38.924 ±0.485	43.785 ±0.758	**40.000** ±0.000	**45.482** ±0.563
P6	50	34.995 ±0.000	37.845 ±0.000	36.785 ±0.925	40.925 ±1.365	37.658 ±0.745	41.896 ±1.125	37.985 ±0.685	42.568 ±0.965	38.456 ±0.568	43.125 ±0.896	**40.000** ±0.000	**45.440** ±0.572
P7	100	69.658 ±0.000	79.324 ±0.000	72.456 ±1.856	82.156 ±2.156	74.258 ±1.425	83.785 ±1.896	75.325 ±1.365	84.568 ±1.756	76.125 ±1.245	85.236 ±1.658	**78.582** ±1.023	**87.651** ±1.856
P8	100	68.536 ±0.000	78.568 ±0.000	71.258 ±2.125	81.365 ±2.568	73.456 ±1.685	82.896 ±2.125	74.256 ±1.568	83.456 ±1.985	74.958 ±1.456	84.125 ±1.896	**76.620** ±1.995	**85.910** ±2.985
P9	200	132.568 ±0.000	148.652 ±0.000	138.256 ±3.568	154.785 ±4.256	142.365 ±2.896	158.426 ±3.456	144.568 ±2.685	160.896 ±3.358	146.125 ±2.358	162.358 ±3.125	**150.630** ±3.338	**169.630** ±5.684
P10	500	335.746 ±0.000	375.685 ±0.000	348.256 ±8.456	388.456 ±9.856	356.785 ±6.785	394.125 ±8.125	362.456 ±6.256	396.785 ±7.856	365.896 ±5.896	398.256 ±7.456	**378.630** ±9.658	**403.890** ±13.582

TABLE IV
TOTAL NUMBER OF DISTINCT OPTIMAL SOLUTIONS OBTAINED BY ALL COMPARED ALGORITHMS OVER 30 INDEPENDENT RUNS, WHERE THE BEST RESULTS FOR EACH TEST INSTANCE ARE HIGHLIGHTED WITH BOLD FONT.

Problems	N_T	HA		ALNS		GA-SWTA		NMA		FDLS-ADE		KCEA	
		Single	Double	Single	Double	Single	Double	Single	Double	Single	Double	Single	Double
P1	10	1	1	3	5	2	3	6	14	5	12	**8**	**23**
P2	20	1	1	5	8	3	5	9	20	8	18	**12**	**33**
P3	24	1	0	8	12	5	8	16	38	15	35	**53**	**102**
P4	24	1	0	7	11	4	7	14	35	12	32	**41**	**108**
P5	50	0	0	12	8	6	3	22	41	25	45	**268**	**586**
P6	50	0	0	10	6	5	2	24	48	28	52	**331**	**605**
P7	100	0	0	5	2	2	1	15	25	18	28	**1205**	**2033**
P8	100	0	0	3	1	1	0	12	19	15	22	**994**	**2248**

lutions for single scenarios). However, this advantage reverses on large-scale instances where FDLS-ADE maintains diversity more effectively. Despite these differences, both algorithms remain substantially inferior to KCEA, particularly evident in large-scale problems.

To illustrate the practical significance of KCEA's diversity preservation capability, Figure 5 presents six distinct optimal solutions discovered for problem instance **P3**. Despite achieving identical total interception probabilities, these solutions exhibit fundamentally different tactical characteristics. The spatial assignment configurations reveal diverse resource allocation patterns, ranging from centralized coordination strategies that concentrate defensive resources around critical areas to distributed approaches that balance coverage across the entire engagement space. The corresponding radar tracking patterns demonstrate the temporal complexity inherent in multimodal MS-SWTA solutions. Some solutions prioritize early target acquisition with extended tracking periods, while others employ just-in-time tracking strategies that minimize radar exposure. These variations reflect different operational philosophies: risk-averse approaches that maximize information gathering versus resource-efficient strategies that minimize signature exposure.

The exponential scaling of solution diversity with problem size represents a critical advantage of KCEA's competitive selection mechanism. For large-scale instances, KCEA identifies over 1000 distinct optimal solutions, providing military planners with unprecedented tactical flexibility. This diversity explosion reflects the algorithm's ability to discover and preserve solutions across multiple modal regions of the solution landscape, each representing fundamentally different resource allocation strategies.

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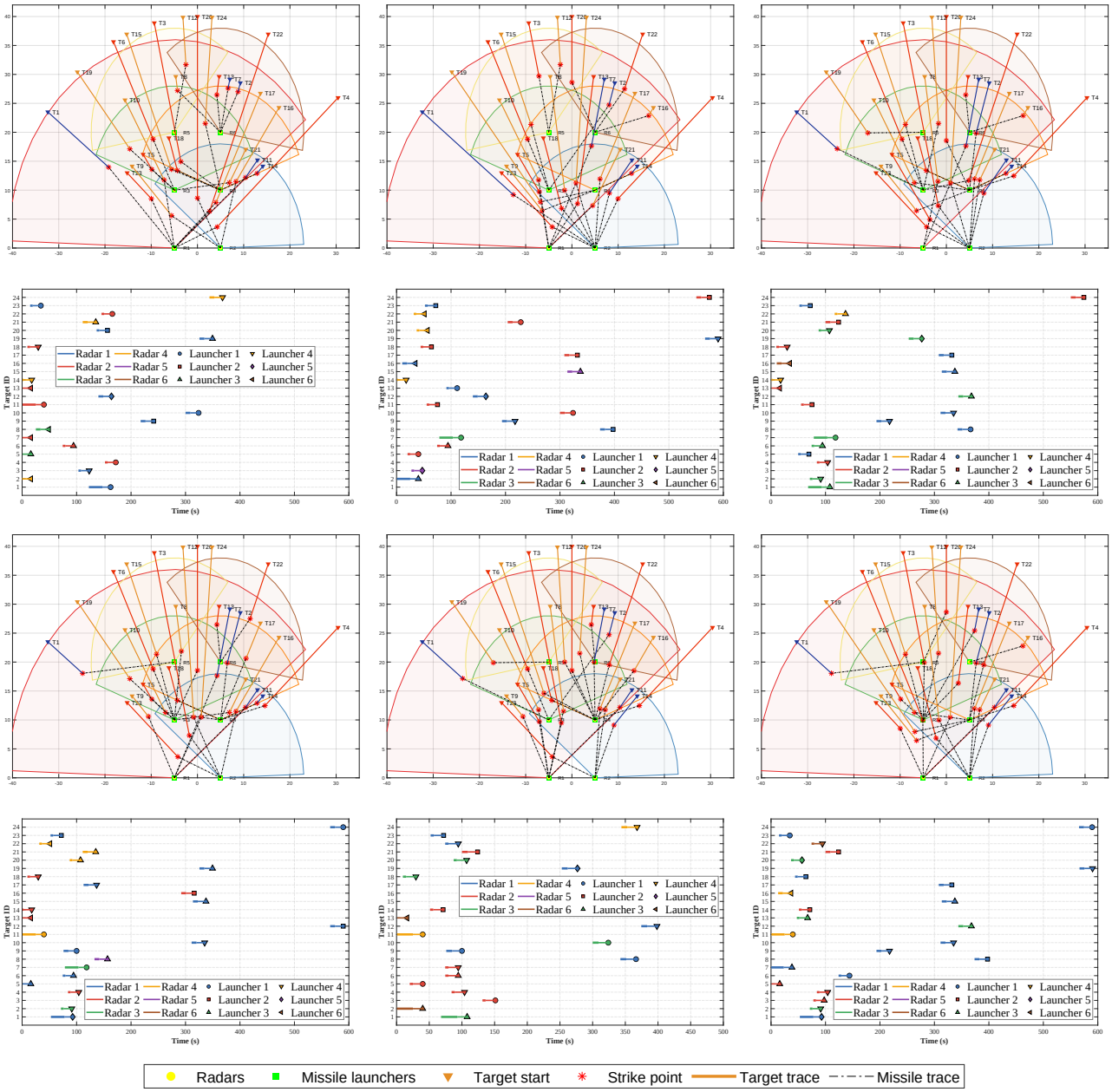


Fig. 5. Six distinct optimal solutions discovered by KCEA for problem instance **P3**, displaying spatial assignment configurations and corresponding radar tracking patterns. Each solution achieves identical interception probability while employing fundamentally different sensor-weapon coordination strategies.

C. Effect of the Proposed Strategies

To validate the effectiveness of individual components within KCEA, we conduct ablation studies using three algorithm variants. KCEA-v1 eliminates the knowledge-guided mechanism, KCEA-v2 removes the competitive selection strategy, and KCEA-v3 represents a standard evolutionary algorithm using only the proposed encoding scheme. Single interception scenarios are employed to isolate the impact of each strategic component.

Table V reveals that total interception probability degradation occurs progressively as key components are removed, with the most significant impact observed for large-scale instances where constraint complexity intensifies.

The knowledge-guided mechanism's contribution becomes evident through KCEA-v1 performance analysis. Without constraint violation learning, KCEA-v1 achieves comparable total interception probabilities to the full algorithm on small-scale problems but shows noticeable degradation for medium and large-scale instances. The competitive selection strategy's importance is demonstrated through KCEA-v2 results, which show the most significant performance degradation among all variants. Removing similarity-based competition results in total interception probability drops of 8.6% for P10, indicating that diversity-aware selection is crucial for maintaining population quality in multimodal landscapes. The standard fitness-based selection in KCEA-v2 leads to premature convergence,

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TABLE V
AVERAGE TOTAL INTERCEPTION PROBABILITY OBTAINED BY KCEA VARIANTS OVER 30 INDEPENDENT RUNS FOR SINGLE INTERCEPTION SCENARIOS.

Problems	N_T	KCEA-v1	KCEA-v2	KCEA-v3	KCEA
P1	10	8.000 ±0.000	8.000 ±0.000	8.000 ±0.000	**8.000** ±0.000
P2	20	16.000 ±0.000	16.000 ±0.000	16.000 ±0.000	**16.000** ±0.000
P3	24	19.200 ±0.000	19.200 ±0.000	19.080 ±0.125	**19.200** ±0.000
P4	24	19.200 ±0.000	19.200 ±0.000	19.025 ±0.158	**19.200** ±0.000
P5	50	38.456 ±0.856	37.825 ±1.256	36.258 ±1.568	**40.000** ±0.000
P6	50	38.125 ±0.925	37.456 ±1.356	35.895 ±1.785	**40.000** ±0.000
P7	100	74.256 ±2.156	71.358 ±2.856	68.456 ±3.256	**78.582** ±1.023
P8	100	72.158 ±2.456	69.856 ±3.125	66.258 ±3.856	**76.620** ±1.995
P9	200	142.856 ±4.856	138.456 ±5.856	131.258 ±6.258	**150.630** ±3.338
P10	500	358.256 ±12.856	345.856 ±15.256	325.456 ±18.856	**378.630** ±9.658

TABLE VI
TOTAL NUMBER OF DISTINCT OPTIMAL SOLUTIONS OBTAINED BY KCEA VARIANTS OVER 30 INDEPENDENT RUNS FOR SINGLE INTERCEPTION SCENARIOS.

Problems	N_T	KCEA-v1	KCEA-v2	KCEA-v3	KCEA
P1	10	6	2	3	**8**
P2	20	8	3	5	**12**
P3	24	25	8	12	**53**
P4	24	22	6	10	**41**
P5	50	85	15	28	**268**
P6	50	92	18	35	**331**
P7	100	245	28	65	**1205**
P8	100	198	22	52	**994**

limiting the algorithm's ability to discover high-quality solutions across different modal regions.

KCEA-v3 represents the baseline evolutionary approach and exhibits the poorest performance across all test instances. The substantial performance gap between KCEA-v3 and the full algorithm underscores the synergistic effect of combining knowledge extraction with competitive selection mechanisms.

The diversity analysis in Table VI provides additional insights into component effectiveness. KCEA-v2 shows the most dramatic diversity reduction, maintaining only 28 distinct solutions for **P7** compared to KCEA's 1205 solutions. This reduction confirms that competitive selection is the primary driver of diversity preservation in multimodal MS-SWTA optimization. KCEA-v1 maintains moderate diversity levels. This suggests that while knowledge-guided operators contribute to diversity maintenance, their primary benefit lies in solution quality improvement rather than multimodal exploration.

The variance analysis reveals that component removal consistently increases solution variability, indicating reduced algorithmic robustness. KCEA maintains the lowest variance across most instances, demonstrating that the integration of knowledge guidance and competitive selection enhances both solution quality and convergence stability.

VI. CONCLUSION

This paper presents a novel Knowledge-guided Competitive Evolutionary Algorithm (KCEA) for Multi-Solution Sensor-Weapon-Target Assignment (MS-SWTA) problems in modern air defense systems. Comprehensive experimental validation on the MADRA-LIB benchmark suite shows that KCEA significantly outperforms state-of-the-art methods across multiple performance dimensions. The algorithm discovers multiple distinct tactical alternatives that represent fundamentally different resource allocation strategies. These strategies range from centralized coordination approaches to distributed coverage patterns. Each strategy suits different operational contexts and threat scenarios. While KCEA is developed specifically for SWTA optimization, it can be extended to solve other combinatorial problems. The competitive selection mechanism provides a general framework for multimodal discrete optimization. The conflict-extraction paradigm can be adapted to other resource-constrained combinatorial problems, though such adaptation requires domain-specific analysis of constraint structures. We acknowledge that applying our complete approach to fundamentally different problem structures requires careful domain adaptation rather than direct application.

Several limitations warrant future investigation despite these significant advances. The current formulation assumes deterministic target trajectories and perfect information availability. These assumptions may not reflect realistic operational conditions characterized by uncertainty and incomplete intelligence. Future research should explore the integration of uncertainty quantification and robust optimization techniques. These techniques would handle trajectory prediction errors and sensor detection uncertainties. The temporal coordination model could be enhanced to incorporate more sophisticated missile guidance strategies. Multiple interceptor coordination scenarios should also be considered. The extension to three-dimensional engagement spaces represents another promising avenue for enhancing practical applicability. Heterogeneous weapon systems should also be incorporated. The knowledge extraction mechanism could benefit from machine learning techniques. These techniques would enable more sophisticated pattern recognition and adaptive guidance strategies.

An important direction for future research involves systematic comparison with reinforcement learning approaches. Potential integration with these approaches should also be explored. This would require developing standardized evaluation frameworks. Training data requirements for military scenarios must be addressed. Hybrid methods that combine the multimodal optimization capabilities of evolutionary algorithms with the adaptive learning strengths of reinforcement learning should be investigated.

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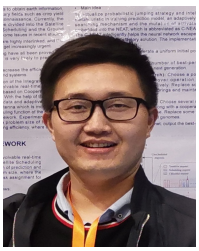


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